Maintenance Schedule

Diesel Engine

EPA/CARB

10V1600A60

12V1600A60

Application Group 5B

MSE50164/01E

Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
10V1600A60	57 kW/cyl.	5B, Continuous operation, variable
12V1600A60	57 kW/cyl.	5B, Continuous operation, variable

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

The maintenance schedule is applicable to products certified in accordance with U.S. emissions regulations and operated in the area of applicability of these regulations.

Information about the latest maintenance schedule is available at: http://www.mtu-online.com.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Nonobservance of the maintenance instructions can result in violation of the statutory emission specifications. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-related components, e.g. control units, sensors, cylinder heads, injectors and exhaust flaps can only be serviced, replaced or repaired if components approved by MTU or equivalent components are used.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability.The maintenance intervals and the scope of required maintenance activities are the result of operational experience and therefore represent recommendations. Severe operating and ambient conditions may require additional maintenance work and/or modification of the maintenance intervals.

Maintenance activities are specified with intervals based on hours in operation and time limits. Whichever value is reached first shall apply.

The individual maintenance intervals are classified in accordance with the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
QL2: Exchange of components and parts (corrective only, not part of the maintenance schedule).
QL3: Maintenance work which requires partial disassembly of the product.
QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

For change intervals for fluids and lubricants as well as for preservation specifications, refer to the relevant instructions of the component manufacturers and the MTU Fluids and Lubricants Specifications. The latest version for drive systems is available online at: http://www.mtu-online.com, and for power generation systems at: http://www.mtuonsiteenergy.com.

Components which are not listed in this maintenance schedule must be serviced in accordance with Manufacturer's Instructions.

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1.2 Allocation matrix application group 5B

<table>
<tr><th colspan="9">10/12V1600C60/A60</th></tr>
<tr><th colspan="2">Load profile</th><th rowspan="2">Load factor %</th><th rowspan="2">Load indicator %</th><th rowspan="2">0 - 60 kW/cyl.</th><th rowspan="2">–</th><th rowspan="2">–</th><th rowspan="2">–</th><th rowspan="2"></th></tr>
<tr><th>Load %</th><th>Time %</th></tr>
<tr><td>110 (100-110)</td><td>–</td><td rowspan="8">0 - 31</td><td rowspan="8">≤ 17</td><td rowspan="8">21,000</td><td rowspan="8">–</td><td rowspan="8">–</td><td rowspan="8">–</td><td rowspan="8">Standard

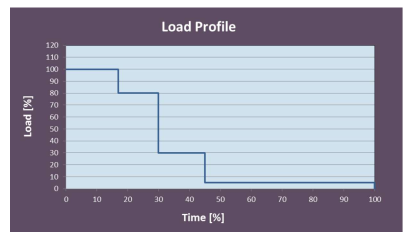
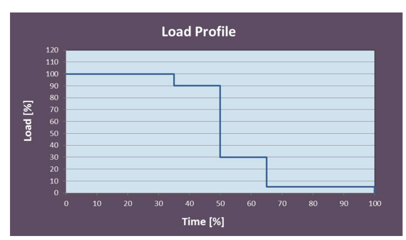
</td></tr>
<tr><td>100 (90-100)</td><td>12</td></tr>
<tr><td>90 (80-90)</td><td>12</td></tr>
<tr><td>80 (65-80)</td><td>–</td></tr>
<tr><td>65 (50-65)</td><td>–</td></tr>
<tr><td>50 (30-50)</td><td>–</td></tr>
<tr><td>30 (5-30)</td><td>16</td></tr>
<tr><td>Idle (0-5)</td><td>60</td></tr>
<tr><td>110 (100-110)</td><td>–</td><td rowspan="8">32 - 35</td><td rowspan="8">≤ 19</td><td rowspan="8">18,000</td><td rowspan="8">–</td><td rowspan="8">–</td><td rowspan="8">–</td><td rowspan="8">
</td></tr>
<tr><td>100 (90-100)</td><td>17</td></tr>
<tr><td>90 (80-90)</td><td>–</td></tr>
<tr><td>80 (65-80)</td><td>13</td></tr>
<tr><td>65 (50-65)</td><td>–</td></tr>
<tr><td>50 (30-50)</td><td>–</td></tr>
<tr><td>30 (5-30)</td><td>15</td></tr>
<tr><td>Idle (0-5)</td><td>55</td></tr>
<tr><td>110 (100-110)</td><td>–</td><td rowspan="8">36 - 45</td><td rowspan="8">≤ 32</td><td rowspan="8">15,000</td><td rowspan="8">–</td><td rowspan="8">–</td><td rowspan="8">–</td><td rowspan="8">
</td></tr>
<tr><td>100 (90-100)</td><td>26</td></tr>
<tr><td>90 (80-90)</td><td>15</td></tr>
<tr><td>80 (65-80)</td><td>–</td></tr>
<tr><td>65 (50-65)</td><td>–</td></tr>
<tr><td>50 (30-50)</td><td>–</td></tr>
<tr><td>30 (5-30)</td><td>12</td></tr>
<tr><td>Idle (0-5)</td><td>47</td></tr>
<tr><td>110 (100-110)</td><td>–</td><td rowspan="8">46 - 55</td><td rowspan="8">≤ 41</td><td rowspan="8">15,000</td><td rowspan="8">–</td><td rowspan="8">–</td><td rowspan="8">–</td><td rowspan="8">
</td></tr>
<tr><td>100 (90-100)</td><td>35</td></tr>
<tr><td>90 (80-90)</td><td>15</td></tr>
<tr><td>80 (65-80)</td><td>–</td></tr>
<tr><td>65 (50-65)</td><td>–</td></tr>
<tr><td>50 (30-50)</td><td>–</td></tr>
<tr><td>30 (5-30)</td><td>15</td></tr>
<tr><td>Idle (0-5)</td><td>35</td></tr>
</table>

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Table 2: Allocation matrix application group 5B

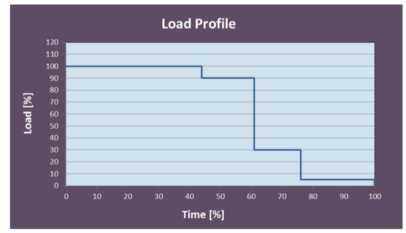
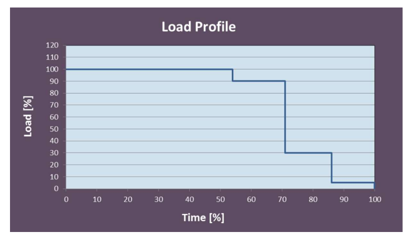
10/12V1600C60/A60								
Load profile		Load factor %	Load indicator %	0 - 60 kW/cyl.	|	|	|	
Load %	Time %							
110 (100–110)	–	56 - 65	≤ 51	12,000	–	–	–	
100 (90–100)	44							
90 (80–90)	17							
80 (65–80)	–							
65 (50–65)	–							
50 (30–50)	–							
30 (5–30)	15							
Idle (0–5)	24							
110 (100–110)	–	66 - 75	≤ 61	9,000	–	–	–	
100 (90–100)	54							
90 (80–90)	17							
80 (65–80)	–							
65 (50–65)	–							
50 (30–50)	–							
30 (5–30)	15							
Idle (0–5)	14							

Table 3: Allocation matrix application group 5B

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1.3 Maintenance schedule matrix 21,000 h

0-21,000 Operating hours

Task	Limit	Operating hours [h]																							
		Daily	1,000	2,000	3,000	4,000	5,000	5,250	6,000	7,000	8,000	9,000	10,000	10,500	11,000	12,000	13,000	14,000	15,000	15,750	16,000	17,000	18,000	19,000	20,000	21,000
Engine																										
W0800	-																									
W1008	2 a																									
W0500	-	X																								
W0501	-	X																								
W0506	-	X																								
W0507	-	X																								
W1001	2 a		X	X	X	X	X		X	X	X	X	X		X	X	X	X	X		X	X	X	X	X	X
W1675	2 a		X	X	X	X	X		X	X	X	X	X		X	X	X	X	X		X	X	X	X	X	X
W1019	-			X		X			X		X		X			X		X			X		X		X	
W1716	-				X				X			X				X			X				X			X
W1207	2 a		X		X				X			X				X			X				X			X
W1003	2 a				X				X			X				X			X				X			X
W8638	-							X						X						X						X
W1525	6 a							X						X						X						X
W1526	6 a							X						X						X						X
W1006	9 a									X								X								X
W1636	9 a									X								X								X
W1298	9 a									X								X								X
W1178	9 a									X								X								X
W1660	-													X												X
W1812	5 a													X												X
W8637	6 a													X												X
W1055	6 a													X												X
W1041	9 a													X												X
W1672	9 a																									X
W3148	18 a																									X

w = weeks
m = months
a = years

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